# Graph-wide discovery capacity and admission gap

Planning assessment, 2026-10-22. Derived inventory and archived observations only; no new scientific trial.

## Fixed inventory and open quota

The prospective P29/P37 schedule has 2,049 and 257 vertices. Three native bases on every vertex give **6,918 mandatory discovery arms** and **51,935,232 target probes**: P29 has 6,147 arms at 8,192 targets each; P37 has 771 arms at 2,048 each. Check indices add 257 P29 and 65 P37 vertices. If `power_x`, transported, and `u=2` arms each receive the full discovery quota, the conditional inventory is **7,884 arms** and **58,650,624 probes**. The draft does not explicitly fix those check-arm quotas; the Coordinator must do so before admission.

## Three archived 32,768-target observations

P29 E2 native (`RUN-ECTD-c769f1`) recorded 68.37 s wall; P29 E3 native (`RUN-ECTD-f395a9`) 77.68 s; P37 floor `u=2` native (`RUN-ECTD-26d06d`) 952.04 s. These are different curves and bases in a separate 20-arm study. The graph shows their measured collector, count, replay, and interval phase times in seconds. Pair-table setup and target probing were not separated, so no full-scan runtime is inferred.




Figure 1: Archived phase wall times on a logarithmic seconds axis. Three output-validated records at 32,768 targets each; run IDs are shown under the marks.

## Next gate

Freeze the check-arm quotas and implementation/seed commitments, then obtain a runnable Coordinator-approved collector contract. The configured Coordinator backend remains unavailable in the current adapter probe. The graph's native yield, useful rank, and full same-point ECDLP costs remain unmeasured. Source paths, exact derived counts, receipt values, and limits are in the accompanying Markdown report and data file.